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Untangling the impact of socio-demographic factors on energy consumption: Why is energy access difficult to achieve in off-grid island communities?

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ABSTRACT

The provision of electricity to rural off-grid island communities is considered a practical initial approach to achieving Just Energy Transition. Often, these communities are powered by diesel generators running on conventional fossil fuels that typically translate to higher electricity tariffs. The use of renewable energy technologies is deemed as the practical approach to electrify these islands. However, there is a need to define how these off-grid communities consume electricity to provide a more robust approach to load demand profiling. This study proposes an alternative path analysis approach to determine causality relationships between socio-demographic attributes of households and their ownership of electrical appliances, as an approach to ascertain electricity consumption. The approach is applied in three island communities in Palawan, Philippines where electricity is provided by diesel generators. A comprehensive survey considering age, gender, income, education, household size, and employment status was done in the three island communities. Path analysis using the R lavaan package was completed to determine causality and the results were compared with traditional correlation analysis. Results of both path analysis and correlation analysis show that income and education have the strongest significance in determining household ability to own appliances, and thus electricity consumption while age is not statistically significant at all. All other factors show weak to moderate significance. The results of the research provide a critical baseline approach to load demand profiling that can help stakeholders to conduct more efficient energy planning and develop more pragmatic strategies towards 100% household electrification and Just Energy Transition.
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Introduction

Electrification has been cogitated to create transformational opportunities for the global population. But while electricity has advanced human lives around the world by enabling the provision of basic vital services such as health care, education, and livelihood, it has also caused severe climate change due to the use of conventional fossil fuels for power generation. The transformation of the global electricity landscape has long been outstanding, with climate change effects such as global warming, severe storms, rising oceans, food insecurity, health risks, and poverty and displacement becoming increasingly rampant. The concept of a just transition from conventional fossil fuels has an extensive history dating back to the early 1970s but has only gained traction through scholarly debates in the 2000s (Wang & Lo, 2021). More recently, the Just Energy Transition (JET) agenda has been coined – a move from dirty and harmful fossil fuels towards a more sustainable, low carbon, and equitable energy system for all. With this agenda, the use of renewable energy technologies (RETs) for electricity generation has gained significant interest worldwide. RETs are considered environmentally friendly and cost-effective because they have been proven to solve the current problem of global warming and environmental degradation (Erdiwansyah et al., 2019; Mahjabeen et al., 2020) while also addressing the economic impact brought about by oil price fluctuations (Abu-Rumman et al., 2020).

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JET is seen as a suitable agenda in the context of energy access and rural electrification, where electricity access is planned for the last-mile populations living in isolated communities (Barone et al., 2021). These isolated communities typically have no or very limited access to electricity, usually sourced from diesel generator sets running on conventional fossil fuels. RETs in these communities prove to be a promising approach to achieving universal access for all while also promoting productivity growth, poverty alleviation, and overall sustainable development (Chen et al., 2020; Dawit Diriba Guta, 2020). Off-grid island communities benefit from off-grid microgrid RET installations (Cabrera et al., 2018), as these systems provide them with increased electricity access at a lower tariff (Chartier et al., 2022; Hirsch et al., 2018; Lozano et al., 2019). However, the electrification of such islandic systems is still hampered by community isolation and socio-economic conditions that affect the technical and economic viability of RETs (Cross et al., 2017). The successful adoption of RETs is still confronted with several challenges. RET implementations in energy access projects typically suffer from insufficient system sizing as a result of uncertainties and inaccurate demand projections, subsequently leading to insufficient supply (Almeshqab & Ustun, 2019; Balderrama et al., 2019; Stevanato et al., 2020). High capitalization requirements of RET implementations coupled with low-income consumers also tend to mask the true cost of electricity, leading to financial failures of projects (Hong & Abe, 2012; Surroop et al., 2018). Moreover, the sustainability of rural electrification initiatives is seen to be highly dependent on socio-economic, demographic, and institutional factors, also taking into consideration the level of acceptance of the users of the technology as well as the value perceived by these consumers of the RETs (Aklin et al., 2018; Mishra & Behera, 2016). For RET projects to be successfully integrated into these small off-grid island communities, there is a need to understand better how the socio-demographic profiles influence electricity consumption and subsequently sustainable energy access as a whole.

In the Philippines, the goal of the Philippine Department of Energy (DOE) is to provide basic electricity access for all Filipinos, as one of the collective long-term visions of the government is termed *Ambisyon 2040* (Energy, n.d.). As of 2020, over 1.26 million Filipino households remain unelectrified, with most of these households living in small off-grid island communities (Energy, 2021). Connecting these small off-grid islands to the main transmission grid of the country is considered economically unviable, thus, electrification is typically achieved through the installation of diesel generators, which usually translates to high electricity tariffs (Lozano et al., 2019) and results in very low electricity access and electricity poverty (Lozano & Taboada, 2020). The use of RETs is deemed to address the concern on costs and electricity deprivation in these small off-grid island communities in the country, helping the country also to achieve Just Energy Transition. However, effective and efficient energy planning requires a better understanding of the demographic and socio-economic characteristics of these small island communities as the success of electrification initiatives is highly dependent on user attributes and perspectives, and how these users consume electricity (da Silveira Bezerra et al., 2017; Hirmer & Cruickshank, 2014). This paper investigates the socio-demographic characteristics of selected off-grid island communities in the Philippines to determine how these affect electricity consumption. Taking into consideration household size, age level distribution, gender ratio, level of education, employment status, and income status of the people living in the island communities, the paper presents an alternative path analysis approach where electricity consumption is considered to be influenced by socio-demographic factors (indirectly) and electrical appliance ownership (directly). A path analysis is done to determine how socio-demographic factors impact household electricity consumption in these islands through their influence on electrical appliance ownership. This analysis is determined to provide a baseline approach that will lead to better electrical load projections, taking into consideration household attributes that will impact appliance ownership and thus electricity consumption. The result of this paper is beneficial to government agencies, private sectors, and other energy stakeholders by providing a better understanding of small off-grid island communities and how they consume electricity, leading to more efficient energy planning and promotion of energy access and just energy transition in islandic communities.

Literature review

Household energy consumption has been identified as one of the major factors contributing to negative environmental impacts and climate change (Dubois et al., 2019; Goldstein et al., 2020; Malla, 2013), attributing to >70 % of carbon footprint globally (Hertwich & Peters, 2009). As of 2019, 13 % of the global population live without electricity access. A majority of these live in impoverished and developing countries (Ritchie et al., 2020), where the grid connection is deemed unviable, and the most practical approach to energy access is through decentralized systems. The provision of electricity through clean and renewable energy sources to these unelectrified households is seen to combat the negative effects of climate change. The electrification of these last-mile households is the most sensible initial approach to achieving Just Energy Transition RETs are the most pragmatic electrification solution. This is especially true in small island developing states and small island communities, where decentralized grids are practical and renewable energy sources are abundant (Atteridge & Savvidou, 2019; Dornan, 2014; Dornan, 2015). The sustainable electrification of these small island communities is highly dependent on their unique socio-demographic attributes that affect the electricity consumption of the households (Kostakis, 2020; Sario, 2018).

While socio-demographic attributes have been extensively studied in relation to electricity consumption, these studies hardly use this causality to predict energy demand. Rather, socio-demographic attributes are most often analyzed in terms of their correlation to certain energy elements. In rural communities, gender, wealth, age, income, educational levels, and household size are used to determine the households' probability of adopting renewable energy technologies and becoming more energy-efficient consumers (Adjakloe et al., 2021; Dawit Diriba Guta, 2020; Guta, 2018; Hyun et al., 2021; Luo et al., 2021). Income and availability of financing schemes are among the socio-economic factors that correlate to rural households' willingness to pay for electricity (Ebers Broughel, 2019; Nduka, 2021; Thapa et al., 2021). In most cases, socio-demographic factors are considered key indicators in studies involving energy security services and determining the level of energy access or energy poverty in poor, rural communities (Bouzarovski & Petrova, 2015; Musango et al., 2020; Siksnelyte-Butkiene et al., 2021). Gender, sex, and other socio-demographic attributes are also being explored in energy research to determine how these impact awareness and perception of energy access for rural households (Biresselioglu et al., 2019; Cannon & Chu, 2021).

Typical studies for rural communities and islandic settings look at the impact of renewable energy implementations or electrification, in general, on the demographic and socio-economic factors within a community (Bertheau et al., 2020; Mazzone, 2019; Mboumboue & Njomo, 2016). The effect of renewables and electricity on the improvement of living conditions of poor rural households is usually explored, and correlation analysis is characteristically done. In most cases, electrification, through renewables, is seen to support the socio-economic development of these poor communities, improving the quality of life for the people (Boliko & Ialnazov, 2019; Meyer & Overen, 2021). On the other hand, electrification projects are seen to be more sustainable if economic conditions are initially improved to ensure that rural households have the purchasing power that will allow them to consume more electricity (Riva & Colombo, 2020; Zhang et al., 2019). Socio-demographic attributes have also been used as predictive or causal factors in energy conservation behavior studies that look at the reduction of carbon

emissions and the effectivity of energy efficiency measures (Soltani et al., 2020; Thapar, 2020).

Socio-demographic attributes, such as those considering geographical distribution, age, gender structure, marital status, and other social and economic situations (Patil & Raut, 2012; Rees, 2020) play a very crucial role in electrification initiatives, affecting technical and financial parameters of such projects (Oluoch et al., 2020; Oluoch et al., 2021). Such factors are also seen as contributory to the households' efficiency in terms of energy consumption, affecting the reliability of RET implementations in rural areas (Balest & Stawinoga, 2022; Dawit Diriba Guta, 2020). Throughout the years, there have been much literature regarding energy consumption and the factors surrounding it. It has been said that electricity consumption is directly related to the socio-demographic characteristics that include the household size, age level distribution, sex ratio, level of education, occupation, and monthly income of the consumer (Malla, 2013; Patil & Raut, 2012; Susanti et al., 2015; Zhou, 2016). For the purpose of this paper, socio-demographic factors are categorized into personal characteristics (covering age, gender, marital status, and family size and composition) and socio-economic characteristics (covering educational attainment and household occupation and income). Table 1 summarizes these factors and how they are perceived to contribute to electricity consumption.

Table 1
Socio-demographic factors as contributing factors to electricity consumption.

Socio-demographic factors	Effect on electricity consumption
Personal characteristics	
• Age	• Electricity consumption in households rises with age (Estiri & Zagheni, 2019), particularly when more adolescents are in the family (Brounen et al., 2012; Jones et al., 2015). This is because teenagers frequently use information technology appliances such as mobile phones that consume more energy than any other appliance (Bartusch et al., 2012; Wallis et al., 2016). • It can also be inferred that a large share of electricity is consumed by people aged 60–70 years, as they are usually retired people who stay at home for longer periods of time, utilizing more energy for heating and cooling (Damari & Kissinger, 2018; Zhou & Teng, 2013). • Younger households also tend to exhibit lower consumption, especially during the daytime, when they are not around in the house (Andersen et al., 2021).
• Gender	• Empirical evidence for energy consumption by gender is inconsistent, with results indicating no significant difference observed in energy consumption between males and females (Frederiks et al., 2015). • However, the gender attribute is significant in terms of electrical appliance use, with women recognized as the primary users of household electrical appliances such as washing machines and dryers, thereby signifying the crucial role of women in promoting flexible electrical consumption (Tjørring et al., 2018).
• Marital Status	• Marital status is associated with owning electrical appliances and computer use, hence, related to electricity consumption (Desipri et al., 2014). • Couples with more than one child tend to exhibit lower energy expenditure, thus lower consumption, while couples with no children and single persons exhibit higher energy expenditure (Besagni & Borgarello, 2020).
• Family size and composition	• Family size is also correlated to increased electricity use, with household electricity consumption increasing by 8 % for every additional family member, a strong indicator that it influences household energy consumption (Damari & Kissinger, 2018; Frederiks et al., 2015; Zhou, 2016; Zhou & Teng, 2013). Therefore, household size significantly affects energy consumption (Blázquez et al., 2013; Desipri et al., 2014; Huang, 2015; Huebner et al., 2016; Ndiaye & Gabriel, 2011; Yalcintas & Kaya, 2017; Ye et al., 2018; Zou & Luo, 2019). • There is also a greater chance of having more appliances when there is an additional household member, mainly because more members tend to possess more devices (Karatasou & Santamouris, 2019).
Socio-economic characteristics	
• Educational attainment	• Families with higher education and income tend to consume more electricity (Abrahamse & Steg, 2009; Huang, 2015; Ye et al., 2018; Zhou & Teng, 2013; Zou & Luo, 2019). • The quality of education received is also relevant in terms of energy consumption, with better quality education relating to increased energy consumption (Wang et al., 2020). • There is also a positive correlation between education and efficient energy consumption, with more educated people tend to consume energy more efficiently, helping curb CO_2 emissions (Bello et al., 2021; Ma et al., 2021; Zaman et al., 2021).
• Household occupation and income	• Electricity usage rises with the increasing income of the household because the number of appliances in a home that offer the opportunity to use electricity is a proxy to wealth (Sakah et al., 2019). • Economic growth positively correlates to electricity consumption as increased income levels also translate to the increased purchasing power of the households (Merlin & Chen, 2021). • Wealthier families are likely to reside in larger homes with more climate control systems, lighting fixtures, and electrical appliances (Damari & Kissinger, 2018;

Materials and methods

Research framework

The research framework is shown in Fig. 1. The study considers the socio-demographic factors identified in Table 1 and classifies them into personal and socio-economic characteristics. The primary data is collected through a survey where a closed-ended survey questionnaire was used considering these factors. The survey questionnaire was approved by the University of San Carlos Research Ethics Committee, where certain qualifications of respondents are identified (i.e., of legal age, has provided informed consent, member of the household, etc.). A preliminary survey was conducted for a set of respondents to validate the survey questions, after which the survey was done in the selected island municipalities considering stratified random sampling, stratified according to the *barangay* (village) population of the municipalities. Data obtained were then analyzed using Structural Equation Modeling (SEM) to determine the relationship of the socio-demographic factors to energy consumption. The final model was tested for its validity and reliability, and the results were compared to the traditional linear regression analysis.

Study areas

The study is implemented in three off-grid island communities in Palawan, where electricity access is provided through diesel generator sets using conventional fossil fuels (see Fig. 2). All islands are composed of public buildings (municipal hall, essential health services, churches, schools), commercial establishments (resorts and transient houses, small restaurants, construction hardware, grocery, and dry good stores), and residential dwellings. Electricity on all three islands is managed and operated by Palawan Electric Cooperative (PALECO). Since these are all island communities, they are not connected to the Palawan primary grid. They have their own diesel power station supplied by the National Power Corporation Small Power Utilities Group (NPC-SPUG).

The first island community is Araceli, classified according to Philippine government guidelines as a fourth-class municipality and covers roughly the northern half of Dumaran Island. The second island community is Cuyo, a coastal community found in the Sulu Sea with an elevation of 22.3 m above mean sea level (PhilAtlas, 2019a). The third study area is Balabac Island, located at the westernmost point in the Philippines. It can be reached by boat, which is a three-hour trip from the nearby municipality, Bataraza. The economy varies on different

Table 1 (*continued*)

Socio-demographic factors	Effect on electricity consumption
	McLoughlin et al., 2012). But, they are also the ones who can invest in energy-saving devices (Frederiks et al., 2015). • A bigger housing unit with high socio-economic status is likely to own more appliances that substantially affect electricity consumption (Kim, 2018)

islands. In Araceli, most people rely on fishing, seafood vending, farming, nipa palm selling, and palm mat weaving. In Cuyo, their economy is based on farming, fishing, cashew eco-agri farming, seaweed (lato) farming, and eco-tourism. The livelihood in Balabac is based on traditional agriculture (kaingin system), animal husbandry, fishing, naval transport, and eco-tourism. The site descriptions are summarized below (Table 2).

Materials

The researchers conducted an actual survey in the selected study areas from June to December 2020, which was distributed to residential households without any demographic restrictions. The sample size for the Structural Equation Modeling (SEM) analysis was determined using the rule of thumb (Ahmad et al., 2016), which requires at least ten times larger than the number of the observed variables. Seven (7) variables or paths are directed to the dependent variable; therefore, the required minimum sample is 70. There are 14,180 households for the three selected islands (Authority, 2019), and a random sample size of 1002 is considered statistically adequate.

The instrument method used in data collection is a paper and pencil questionnaire adopted from the previous research (Lozano et al., 2019). The collection of various details on the questionnaire has four (4) parts: (1) personal information; (2) household information; (3) electricity consumption; and (4) future additional capacity. In addition, the respondents were asked to sign a consent form indicating that they were voluntarily taking part in the survey. The results of the survey were presented according to the socio-demographic factors that affect energy consumption. Structural equation modeling (SEM) in R lavaan package was utilized for this study to identify the most significant electricity consumption determinants. Then, to test the accuracy and consistency of the method, the validity and reliability of the model were measured and compared to the results of linear regression analysis.

Data analysis procedure

The use of statistical software for data analysis, specifically the correlation, path, and factor analyses, provides an outstanding result in determining independent variables' association with a dependent variable. Structural Equation Modeling (SEM) has become a more popular approach to multivariate statistical analysis that analyses unobserved latent variables measured by multiple factors while also identifying relationships between them (Đurišić et al., 2020; Kim et al., 2020). This is considered a more robust approach than other multivariate methods (Motawa & Oladokun, 2015) because it combines the best features of confirmatory factor analysis, path analysis, and multiple linear regression analysis into a single tool (Liu et al., 2020). Among the characteristics of SEM that differentiates it from any other modeling method is its ability to incorporate both measurement and latent variables in the same analysis, which results in a more substantial and better assessment of the proposed model (Belaïd, 2017; Lowry & Gaskin, 2014).

In performing SEM, validating the proposed model (measurement model) and fitting the structural model are of great importance. The proposed model, or measurement model, draws the connections among measured variables underlying the latent variables. The structural model describes the relationships between the latent variables that have been identified by the theory (Kursunoglu & Onder, 2019). The proposed models are verified for their suitability using indicators such as Chi-square (x^2), Goodness of Fit Index (GFI), Root Mean Square Error of Approximation (RMSEA), Comparative Fit Index (CFI), Normed Fit Index (NFI), Tucker-Lewis Index (TLI), Adjusted Goodness of Fit Index (AGFI), and chi-square/degree of freedom (x^2/df) (Kim et al., 2020). Table 4 shows the acceptable values for these indicators (Ahmad et al., 2016; Hair, 1998).

Suppose the fit of the proposed model did not pass the level of acceptance, the initial hypotheses regarding measurement may be wrong, which may affect the fit of the structural model (Kursunoglu & Onder, 2019). However, if the measurement fit has an acceptable value, the subsequent step is to examine the fit of the structural model by performing model validation. It will improve the measurement model's reliability and stability, which includes assessing average variance extracted (AVE) and composite reliability (CR). Validity is about the accuracy of the model being measured and the extent to which the results measure, while reliability is the degree of consistency of the model being measured and how reliable it is in measuring intended latent constructs (Zeb, 2011). Here are the assessment criteria that will be used to do this:

a. the factor loadings of each item should be over 0.50 to be significant;
b. the value of the AVE should be higher than 0.50 to be statistically significant; and
c. the CR should be >0.70 to be valid and reliable (Belaïd, 2017; Liu et al., 2020);

The equations written below were used to solve the values of AVE and CR.

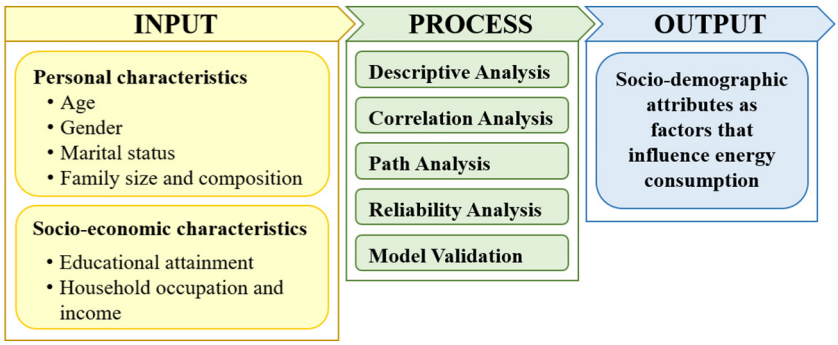


Fig. 1. Research framework.

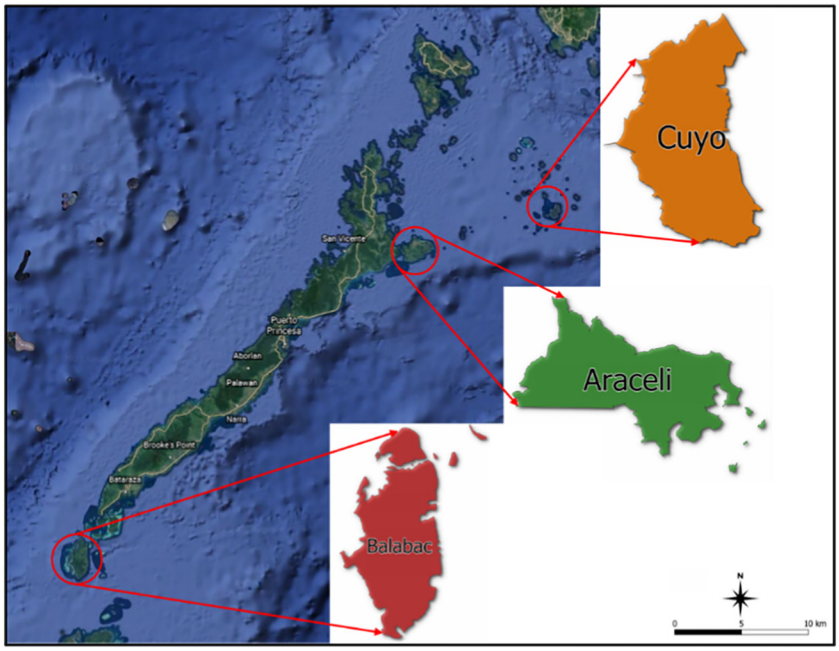


Fig. 2. Map of the selected study areas.

$$AVE = \frac{\sum_{i=1}^{n} \lambda_i^2}{n} \quad (1)$$

$$CR = \frac{\left(\sum_{i=1}^{n} \lambda_i\right)^2}{\left(\sum_{i=1}^{n} \lambda_i\right)^2 + \sum_{c=1}^{n} \delta_i} \quad (2)$$

where λ is the FL of each item, δ is the error variance, and n is the number of items or observed variables (Black et al., 2009).

High reliability may indicate that the measurement model is valid, but it is not enough to ensure validity. However, when the measurement is valid, it is also usually reliable. That is why validity and reliability are closely related to each other, and these two should be done to test the measurement model. SEM can describe both the direct and indirect effects between the associated variables and the overall outcome of direct and indirect effects (Motawa & Oladokun, 2015; Sarstedt et al., 2017). It can answer several interrelated questions in a single, systematic, and comprehensive analysis by modeling the relationships between multiple independent and dependent variables at once (Kline, 2011; Liu et al., 2020). This study will apply the SEM approach in this manner: (1) descriptive analysis; (2) correlation analysis; (3) path analysis; and (4) reliability analysis.

Descriptive analysis

The descriptive analysis provides a summary of the responses from the field survey conducted. It also tells the skewness and kurtosis to verify the normality of univariate distribution. Skewness is a measure of the dataset's symmetry that describes the amount and direction of the skew. The skewness value ranges from −1 to +1, and according to the general rule:

Kurtosis measures both the peak and tail weights of the frequency distribution, and a value of 1.0 means a perfectly normal distribution. Distributions with a value >1.0 indicate positive distribution or leptokurtic distribution, suggesting a high peak. Distributions with a value <1.0 indicate negative distribution called platykurtic distribution, meaning a flat-topped curve (Kline, 2011).

Correlation analysis

The correlation analysis determined the degree of relationship between two or more variables (Kline, 2011), but it does not indicate the causal factors to energy consumption. When the coefficient of correlation (r) is +1, it suggests that two variables are perfectly associated positively, but if r = −1, it means a perfect negative association among the variables. If r = 0, there is no association among the variables (Schober et al., 2018; Senthilnathan, 2019).

Path analysis

The path analysis of the proposed model, or measurement model, emphasizes the direct and indirect relationships of measured or independent variables to the dependent variable. The theoretical model of this study in Fig. 3 shows the direct and indirect effects on energy consumption (Estiri, 2015). The indirect effect represents the socio-demographic factors (i.e., household size, age, gender, level of education, employment status, and income status) that affect the number of electrical appliances owned, which may directly impact energy consumption.

Table 2
Site description of the selected areas (Authority, 2019; Palawan Electric Cooperative, 2020; PhilAtlas, 2019b).

Area name	Location	Area (km^2)	Population	No. of households	Density (population/km^2)	No. of households connected to electricity
Araceli	Latitude: 10°33′32″ N Longitude: 119°59′40″ E	204.30	14,895	3294	73	758
Cuyo	Latitude: 10°51′ N Longitude: 121°01′ E	84.95	34,380	7836	263	6905
Balabac	Latitude: 07°59′ N Longitude: 117°03′ E	581.6	13,759	3050	69	787

Table 3
Sample population per municipality and per barangay.

Municipality	Barangay	Total population (households)	Sample population (households)
Araceli	13	3294	335
Cuyo	28	7836	300
Balabac	14	3050	367

Table 4
Model fit indices.

Category	Model fit index	Level of acceptance
Absolute fit	Chi-square (x^2)	$p > 0.05$
	Root Mean Square Error of Approximation (RMSEA)	<0.08
	Goodness of Fit Index (GFI)	>0.90
Incremental fit	Adjusted Goodness of Fit Index (AGFI)	>0.90
	Comparative Fit Index (CFI)	>0.90
	Tucker-Lewis Index (TLI)	>0.90
	Normed Fit Index (NFI)	>0.90
Parsimonious fit	Chi-square/Degree of Freedom (x^2/df)	<5.0

Fig. 4 shows the graphical representation known as the path diagram to establish the causal relationship among the measured variables to energy consumption. Its goal is to determine the significance of socio-demographic factors and the extent to which they affect energy consumption.

Pearson Correlation Coefficient (parametric) and Spearman Rank Correlation (non-parametric) were used in measuring the strength or extent of association between variables and the direction of the relationship. The probability value (p-value) indicates the significance of the relationship, and for this study, 0.05 is the critical value, where:

$p \leq 0.05$ (statistically significant relationship).

$p > 0.05$ (NOT statistically significant relationship).

Spearman's values (ρ), which measure the strength of the relationship, should have values ranging from -1 to $+1$, with $+$ values which indicate a positive relationship and $-$ values indicating a negative relationship. A correlation coefficient of zero implies that the relationship between variables is weak or has no association at all (Fig. 5) (Gogtay & Thatte, 2017; Senthilnathan, 2019; Weaver et al., 2018).

Reliability analysis

Model fit indices were used for model verification and validation. The model fit indices to be used for judging good fit are the chi-squared test (x^2), the goodness of fit (GFI), RMSEA, CFI, and TLI (Karatasou & Santamouris, 2019). The chi-squared test is the traditional measure to assess overall model fit and the discrepancy between the sample and fitted covariance matrices (Hooper et al., 2008). However, most researchers do not consider this test a very useful fit index because it is sensitive to sample size. For this reason, an alternative measure of fit called normed or relative chi-square can be used by dividing x^2 by degrees of freedom (df), x^2/df, and the value of this is less sensitive to sample size. Different researchers have their own acceptance criteria, ranging from one to less than or equal to five. A more sensible approach that can be used to assess whether the model fits approximately to the population is the RMSEA (Schermelleh-Engel & Moosbrugger, 2003). It is a parsimony-adjusted index, meaning it avoids sample size issues by analyzing the discrepancy between the hypothesized model and the population covariance matrix (Hooper et al., 2008). RMSEA value smaller than 0.05 indicates a convergence fit, values between the range of 0.05 to 0.08 indicate a fit close to good, and values falling between 0.08 and 0.10 is a mediocre fit, but when it is >0.10, it is not acceptable (Cangur & Ercan, 2015). The CFI, a revised form of the Normed Fit Index (NFI), is an incremental fit index that can perform well even with a small sample size (Hooper et al., 2008). It ranges from 0 to 1, with higher values indicating an acceptable fit (Jones et al., 2015), but a cut-off criterion of ≥0.90 closer to 1 means a very good fit (Li & Lai, 2011; Sunbul et al., 2018). TLI or non-normed fit index (NNFI) measures relative fit and was developed in response to the disadvantage of NFI with regards to being affected by sample size (Cangur & Ercan, 2015). TLI values are often very close to 1 (Doloi et al., 2011), but values ≥0.90 are presently accepted as an indicator of a good fit (Li & Lai, 2011; Sunbul et al., 2018). The next step when all measurement fit has acceptable values was performing model validation of the structural model by following the criteria below and using Eqs. (1) & (2). It improved the reliability and stability of the measurement model, including assessing the AVE and CR.

Results & discussions

Descriptive statistics

Table 6 presents the descriptive analysis derived from survey results, considering both categorical and numerical variables. Categorical variables are gender, level of education and employment status, while numerical variables are household size, age, income, number of appliances owned, and monthly energy consumption. For analysis, numerical values are assigned for each categorical variables and for ranges of the numerical variables, as shown in Appendix A. The mean size per household is 3.731, or approximately four persons per dwelling. The residents' educational attainment is mostly elementary or high school graduates, with a monthly income of US $21.00–US $200.00 (PhP1,000.00–10,000.00). Since most parts of the island do not have energy access, the electricity-consuming devices range from 0 to 10 appliances. Among those are ordinary light bulbs, electric fans, phone chargers, rice cookers, and television. Energy sources are either from the local distribution utility (56.7 %), self-owned generators (1.4 %), island generators supplied by the barangay (10.6 %), generators from other sources (0.4 %), or self-owned solar PV (2.3 %). Only 59.6 % have access to electricity, while 40.4 % do not have one. Those who have electricity use it for 4–5 h daily, with energy consumption ranging from 0 to 111 kWh per month. Those who do not have electricity use kerosene lamps, solar light, and candles as lighting sources during the night.

The skewness and kurtosis values are verified to test the normality of univariate distribution with acceptable values ranging from -1 to $+1$ (Table 5). Based on the results, household size, employment status, income status, and energy consumption are right-skewed distribution. Gender and the number of appliances are moderately skewed. The age and level of education have an approximate symmetrical distribution. The Kurtosis values of household size, age, gender, level of education, employment status, and the number of electrical appliances are <1.0,



Fig. 3. A Theoretical model of direct and indirect effects on energy consumption.

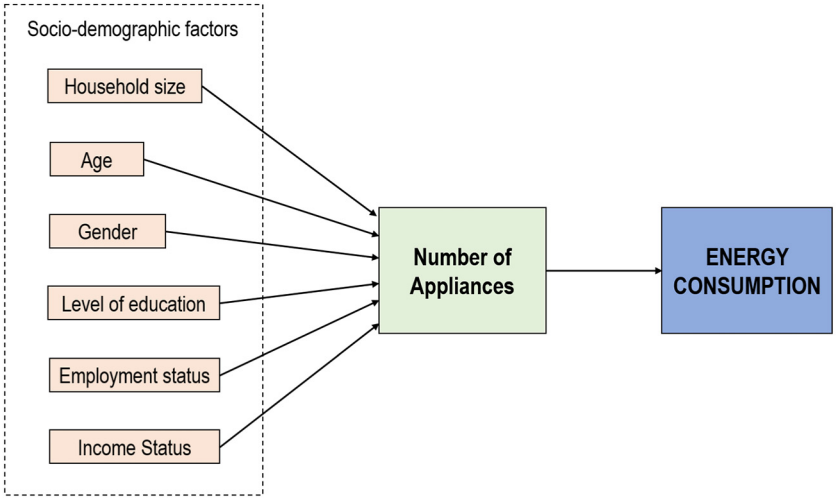


Fig. 4. Initial path design of the measurement model.

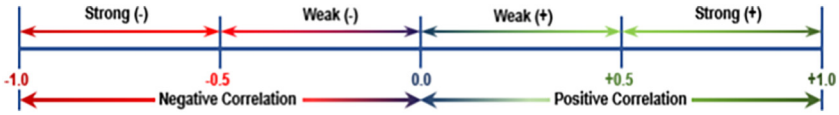


Fig. 5. Correlation coefficient spectrum.

indicating negative or platykurtic distribution. Whereas income status and energy consumption indicate positive or leptokurtic distribution with values >1.0. The distribution plot can be seen in the Fig. 6.

Correlation analysis

The result of a simple correlation analysis of socio-demographic characteristics to number of appliances and energy consumption is in Table 7, while Tables 8 and 9 show the interpretation of the correlation analysis of the different socio-demographic factors to the number of appliances and energy consumption.

Among the socio-demographic factors analyzed, age is determined to be a non-significant factor in influencing number of appliances owned while all the other socio-demographic factors examined shows significant influence to the number of appliances owned. The strongest positive determinants are level of education and income status, signifying that these two factors are directly correlated to the number of appliances owned in a household.

In the analysis of correlationships between the socio-demographic factors and energy consumption, it is seen that age is still a non-significant factor while all the others are significant. Employment status and income status still show the strongest relationship with energy consumption, validating the results shown in Table 8. However, while correlation analysis helps identify the significant relationships between the socio-demographic attributes (independent variables) and energy consumption, it does not imply causation and the assessment of significance is still simple. The path coefficient analysis was performed to clarify these relationships.

Table 5
Skewness.

Value	Skewness
>1	Right-skewed
<−1	Left-skewed
−1 to −0.5	Moderately skewed
0.5 to 1	Moderately skewed
−0.5 to 0.5	Approximately symmetric

Path analysis

The path coefficient analysis was done using SEM in R lavaan package. It clearly shows the causal relationship between the measurement variables. Table 10 and Fig. 7 present the direct and indirect effects of socio-demographic factors on energy consumption. The path coefficient analysis gives a good result and proves the theoretical model shown in Fig. 4 that appliance usage directly affects and socio-demographic characteristics indirectly impact energy consumption.

The number of appliances is statistically significant with energy consumption at a p-level of 0.000 ($p \leq 0.05$) and showed the strength of the relationship at 85.1 %. For the indirect effects, only the age poses no significant association with the number of appliances with a p-value of 0.079 ($p \leq 0.05$), while all the other factors have p-values of 0.000 and 0.007 ($p \leq 0.05$). The path diagram shows that at 36.9 % and 35.2 %, the income status and the educational attainment have the highest indirect effect, suggesting that they are strong predictors of owning more electrical appliances that consume a high amount of electricity. Household size at −19.5 % shows a moderate negative correlation. Gender is also a moderate predictor at 13.9 %, while age (9 %) and employment status (2.2 %) are the weakest predictors among the socio-demographic factors.

Reliability analysis

Model fit indices GFI, RMSEA, CFI, TLI, and x^2/df were verified for model verification and validation. Factor analysis was done to compute the values of AVE and CR to examine the measurement reliability of the model being measured. It has to pass the criteria to prove the validity and reliability of the measurement variables. The result of the model fitness is shown in Table 11.

Both GFI and RMSEA are absolute fit indexes with different ranges of measurement values. GFI varies from zero (no fit) to one (perfect fit),

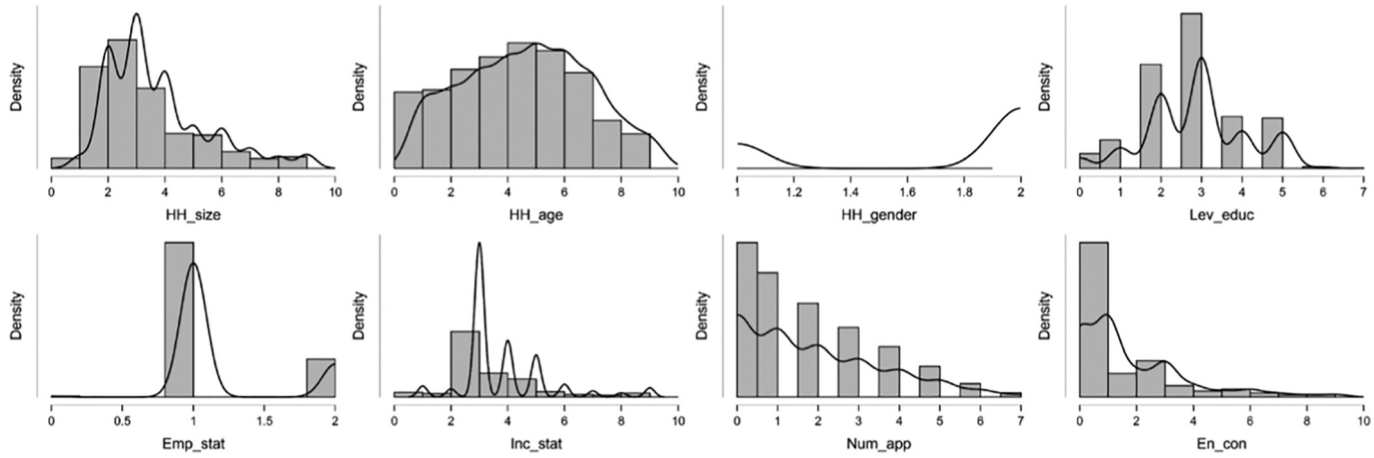


Fig. 6. Distribution plot of measurement variables.

while the RMSEA value indicates a fit close to good between 0.05 and 0.08. Since GFI is 0.983 > 0.90, and RMSEA is 0.056 < 0.08, therefore, it passed the tests and have a good fit.

CFI and TLI are incremental fit indices with values ranging from 0 to 1, but the level of acceptance should be >0.90. The model has significant values of 0.990 and 0.979, respectively, which indicates that the values are closer to 1 and are a perfect fit.

Another alternative measure of fit is normed or relative chi-square, x^2/df, which has a criterion ranging from one to five or <5. The model has 4.147, which shows that it has overall goodness of fit.

Since all measurement fit has acceptable values, the last step is to determine the validity and reliability of the model. AVE and CR are computed to get this. Factor loadings (FL) were determined by using factor analysis in R. Substituting the sum of the square of all loadings ($\lambda^2 = 4.538$) and $n = 8$ to Eq. (1), the value of AVE is solved below.

$$\text{AVE} = \frac{4.538}{8} = 0.567$$

For composite reliability test, use Eq. (2) where $\lambda = 4.826$ and $\delta = 3.462$,

$$\text{CR} = \frac{(4.826)^2}{(4.826)^2 + 3.462} = 0.871$$

Table 12 summarizes the results for the evaluation of the reliability and stability of the measurement model.

The variables should have factor loadings of at least |0.50| to be considered significant as a rule of thumb. All factor loadings are higher than 0.50, except gender having a negative value. The value of AVE is 0.567, which indicates that all items in the measurement model are statistically significant. The composite reliability is 0.871, with a value between 0.70 and 0.90, which is considered satisfactory. Therefore, the measurement is valid and reliable.

Table 6
Measurement variables and descriptive statistics.

Measurement variables	Code	Mean	SD.	Min.	Max.	Skewness	Kurtosis
Household size	HH_size	3.731	1.785	1	9	1.116	0.829
Age	Age	4.672	2.208	1	9	0.035	−0.865
Gender	Gender	1.711	0.454	1	2	−0.930	−1.137
Level of education	Lev_educ	2.887	1.234	0	7	0.011	−0.028
Employment status	Emp_stat	1.192	0.406	0	2	1.343	0.507
Income status (monthly)	Inc_stat	3.819	1.538	1	9	1.445	2.728
Number of electrical appliances	Num_app	1.824	1.719	0	7	0.800	−0.160
Energy consumption	En_con	1.740	1.934	0	9	1.554	2.326

Conclusions and policy implications

This paper aimed to investigate the different socio-demographic characteristics of households in off-grid island communities in Palawan, Philippines and how they affect energy consumption of these households. A comprehensive survey was done to collect information about the household size, gender, level of education, employment status, income status, monthly electricity consumption, and the number of energy-consuming appliances of the residents. Correlation analysis and structural equation modeling (path analysis) were used to determine, respectively, the correlationship and causality of each socio-demographic attribute to ownership of appliances and energy consumption. Comparative analysis (model validity and reliability) were done to evaluate the results of the two analyses, indicating that both have comparable results.

Among the measured variables, income status has the most substantial indirect effect at 36.9 % and is significantly associated with the number of appliances, having a p-value of 0.000 ($p \leq 0.05$). It clearly indicates that it is a strong predictor of owning more electrical appliances, thus higher electricity consumption. Other indirect factors that showed a positive association and statistically significant relationship are the educational level (35.2 %), gender (13.9 %), and employment status (2.2 %), with p-values of 0.000 ($p \leq 0.05$). Age poses less association and contribution and is not a significant factor for this study at a path coefficient value of 9 % and p-value of 0.079 ($p \leq 0.05$). In contrast, household size (−19.5 %) negatively contributes to having more energy-consuming appliances in the household, but a significant factor, nonetheless, having a p-value of 0.000 ($p \leq 0.05$). The number of electrical appliances that directly affect energy consumption has a p-value of 0.000 ($p \leq 0.05$) and a path coefficient value of 85.1 %. It indicates the strength of the relationship and a statistically significant factor in energy consumption. The path coefficient analysis presented satisfactory results, proving that appliance usage directly affects energy consumption, while socio-demographic characteristics indirectly. Several model fit indices were performed, and the results have acceptable values. The overall goodness of fit called normed or relative chi-square, x^2/df, has a value of 4.147; GFI = 0.983; RMSEA = 0.056; CFI = 0.990; and TLI = 0.979. All model fit measurement values passed the tests and are close to a perfect fit. From the result of validity and reliability tests, AVE is 0.567, which is greater than the criteria value of 0.5, which means that all the measurement model items are statistically significant. The reliability of measured variables and internal consistency has

Table 7
Correlation analysis between independent and dependent variables.

	HH_size	Age	Gender	Lev_educ	Emp_stat	Inc_stat	Num_app	En_con
HH_size	1.00							
Age	0.141*	1.00						
Gender	−0.163*	−0.119*	1.00					
Lev_educ	−0.280*	−0.240*	0.197*	1.00				
Emp_stat	0.007	−0.024	0.090*	0.087*	1.00			
Inc_stat	−0.077*	−0.045	0.019	0.357*	0.063*	1.00		
Num_app	−0.332*	−0.055	0.238*	0.546*	0.085*	0.510*	1.00	
En_con	−0.274*	−0.041	0.217*	0.470*	0.127*	0.485*	0.851*	1.00

* Indicates significance at the level of 5 %.

Table 8
Interpretation of the correlation analysis between socio-demographic factors to the number of appliances.

Factors	Correlation		Strength of relationship			
	Significant	Not significant	Weak (−)	Strong (−)	Weak (+)	Strong (+)
HH_size	Yes	–	Yes	–	–	–
Age	–	Yes	Yes	–	–	–
Gender	Yes	–	–	–	Yes	–
Lev_educ	Yes	–	–	–	–	Yes
Emp_stat	Yes	–	–	–	Yes	–
Inc_stat	Yes	–	–	–	–	Yes

Table 9
Interpretation of the correlation analysis between socio-demographic factors to the energy consumption.

Factors	Correlation		Strength of relationship			
	Significant	Not significant	Weak (−)	Strong (−)	Weak (+)	Strong (+)
HH_size	Yes	–	Yes	–	–	–
Age	–	Yes	Yes	–	–	–
Gender	Yes	–	–	–	Yes	–
Lev_educ	Yes	–	–	–	–	Yes
Emp_stat	Yes	–	–	–	Yes	–
Inc_stat	Yes	–	–	–	–	Yes
Num_app	Yes	–	–	–	–	Yes

a required value of CR ≥ 0.6, and from the result, CR = 0.871. Therefore, the measurement model is valid and reliable.

Energy access in off-grid remote island communities is yet to be improved and the use of renewable energy technologies is considered the most pragmatic solution in providing increased electricity access to these isolated communities. The projection of potential load demand in these off-grid communities proves to be a challenge because of lack of historical data. This leads to ineffective load demand profiling, inaccurate system sizing, and ultimately unsustainable energy access projects. The results of this study can help provide a more vigorous approach to load demand profiling by considering the socio-demographic attributes of households in these off-grid communities. This could serve as baseline data for effective energy planning that could help the Philippines strategize for its transition to cleaner and more sustainable energy sources. Moreover, a more robust characterisation of electricity demand could help policy-makers and other stakeholders make more pragmatic programs that will address the goal of 100 % household electrification and Just Energy Transition in the Philippines. Likewise, the approach used in the study could be used in other regions, especially in small island developing states across the globe. The researchers recommend potential follow-up studies grounded by the results of this research such as resource mapping of renewable energy resources and system design optimization based on load demand profiles considering socio-demographic attributes.

Table 10
Structural modeling analysis results of the substantive model.

Causal relationship	p-Value	Path coefficient (ρ)
En_con ← Num_app	0.000	0.851
Num_app ← HH_size	0.000	−0.195
Num_app ← Age	0.079	0.090
Num_app ← Gender	0.000	0.139
Num_app ← Lev_educ	0.000	0.352
Num_app ← Emp_stat	0.007	0.022
Num_app ← Inc_stat	0.000	0.369

Table 11
Model fit result of the measurement model.

Model fit indices	Level of acceptance	Substantive model
GFI	>0.90	0.983
RMSEA	<0.08	0.056
CFI	>0.90	0.990
TLI	>0.90	0.979
x^2/df	<5.0	4.147

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Institutional review board statement

The study was conducted with the approval of the University of San Carlos Research Ethics Committee (Protocol Number 084/2019–11-supapo and the Approval Date December 12, 2019).

Informed consent statement

Informed consent was obtained from all subjects involved in the study.

Table 12
Measurement validity results.

Item	Factor loading (FL)	AVE	CR
HH_size	0.584	0.567	0.871
Age	0.724		
Gender	−0.547		
Lev_educ	0.612		
Emp_stat	0.925		
Inc_stat	0.750		
Num_app	0.900		
En_con	0.878		

Note: Applied rotation method is varimax.

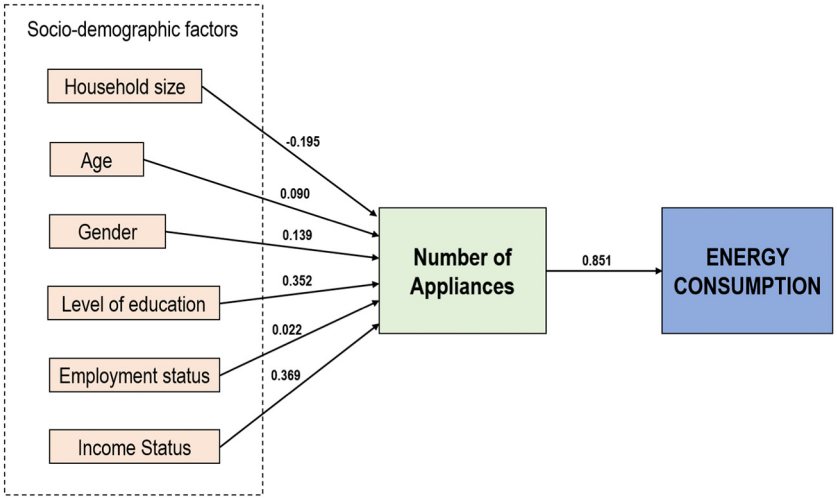


Fig. 7. Path diagram of the measurement model.

Data availability statement

Data is available upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Measurement variables

Measurement variables	Code	Scale
Household size	HH_size	1 = 1 person 2 = 2 persons 3 = 3 persons 4 = 4 persons 5 = 5 persons 6 = 6 persons 7 = 7 persons 8 = 8 persons 9 = 9 persons or more
Age	Age	1 = 18 to 23-year-old 2 = 24 to 29-year-old 3 = 30 to 35-year-old 4 = 36 to 41-year-old 5 = 42 to 47-year-old 6 = 48 to 53-year-old 7 = 54 to 59-year-old 8 = 60 to 65-year-old 9 = over 65-year-old
Gender	Gender	1 = male 2 = female
Level of education	Lev_educ	0 = No formal schooling 1 = Did not graduate elementary level 2 = Elementary graduate 3 = High school graduate 4 = College level but did not graduate 5 = Bachelor's degree 6 = Master's degree 7 = Doctoral degree 8 = Others
Employment status	Emp_stat	0 = Not employed 1 = Employed 2 = With business
Income status (monthly)	Inc_stat	1 = PhP0–500.00 (US $0–10.00)* 2 = PhP501.00–1000.00 (US $11.00–20.00)* 3 = PhP1001.00–5000.00 (US $21.00–100.00)* 4 = PhP5001.00–10,000.00 (US $101.00–200.00)* 5 = PhP10,001.00–15,000.00 (US $201.00–300.00)* 6 = PhP15,001.00–20,000.00 (US $301.00–400.00)* 7 = PhP20,001.00–25,000.00 (US $401.00–500.00)* 8 = PhP25,001.00–30,000.00 (US $501.00–600.00)* 9 = above PhP30,000.00 (above US $600.00)*
Number of electrical appliances	Num_app	0 = None 1 = 1–5 appliances 2 = 6–10 appliances 3 = 11–15 appliances 4 = 16–20 appliances 5 = 21–25 appliances 6 = 26–30 appliances 7 = 31–35 appliances 8 = 36–40 appliances 9 = more than 40 appliances
Energy consumption	En_con	0 = None 1 = 1–111 kWh 2 = 112–223 kWh 3 = 224–335 kWh 4 = 336–447 kWh 5 = 448–559 kWh 6 = 560–671 kWh 7 = 672–783 kWh 8 = 784–895 kWh 9 = above 895 kWh

Note: *Exchange rate is based on PhP50.00 = US $1.00.

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